Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech
Code: 110501

Subject: Analog & Digital Communication System

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which modulation technique transmits one complete sideband and a part of another sideband
- (i) AM
- (ii) DSBSC
- (iii) SSB-SC
- (iv) VSB-SC

(b) Which of the following is an advantage of FM over AM?
- (i) Lower bandwidth requirement
- (ii) Simpler transmitter
- (iii) Better noise immunity
- (iv) Lower transmission quality

(c) According to the Sampling Theorem, the sampling frequency should be
- (i) Equal to signal frequency
- (ii) At least twice the highest frequency of the signal
- (iii) Less than signal frequency
- (iv) Half of the signal frequency

(d) What is the primary purpose of error control coding?
- (i) Increase bandwidth
- (ii) Reduce transmission speed
- (iii) Compress data
- (iv) Detect and correct errors in transmitted data

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(e) A major advantage of optical communication is
- (i) High data transmission rate
- (ii) Low bandwidth capacity
- (iii) High signal attenuation
- (iv) High electromagnetic interference

(f) Which frequency range is generally used in microwave communication systems?
- (i) Audio frequency range
- (ii) GHz frequency range
- (iii) Radio frequency below 1 MHz
- (iv) Infrared frequency range

(g) In cyclic codes, the received code word is checked for errors using
- (i) Syndrome calculation
- (ii) Modulation
- (iii) Multiplication
- (iv) Sampling

(h) Which pulse modulation technique changes the position of the pulse according to the message signal?
- (i) PAM
- (ii) PWM
- (iii) PCM
- (iv) PPM

(i) Which modulation technique belongs to angle modulation?
- (i) AM
- (ii) DSBSC
- (iii) FM
- (iv) VSB-SC

(j) What is the full form of DSBSC?
- (i) Double Signal Band Suppressed Carrier
- (ii) Dual Side Band Signal Carrier
- (iii) Double Side Band Suppressed Carrier
- (iv) Digital Side Band Suppressed Carrier

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Q.2 (a) Explain the basic blocks of a communication system with a neat block diagram and describe the function of each block. [7]

(b) Explain the principle of SSB-SC modulation. Discuss its advantages and disadvantages over conventional AM. [7]

Q.3 (a) Explain the concept of angle modulation and differentiate between Frequency Modulation and Phase Modulation. [7]

(b) Explain the transmission bandwidth of FM signals and derive Carson's Rule. [7]

Q.4 (a) Explain Pulse Amplitude Modulation with suitable waveform diagrams. [7]

(b) Compare PAM, PWM, PPM, and PCM. [7]

Q.5 (a) Compare linear block codes, cyclic codes, and convolution codes. [7]

(b) Discuss the advantages and limitations of convolution coding in digital communication systems. [7]

Q.6 (a) Explain the working principle of microwave communication systems with suitable block diagrams. [7]

(b) Describe the architecture and working principle of a mobile communication system. [7]

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Q.7 (a) Compare AM, DSBSC, SSB-SC, and VSB-SC modulation techniques. [7]

(b) Explain the working of an FM transmitter with a neat block diagram. [7]

Q.8 (a) Describe the principle of Phase Modulation and derive the expression for a PM signal. [7]

(b) Discuss the generation and characteristics of DSBSC modulation with suitable diagrams. [7]

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Q.9 Write short note on *any two* of the following: [7 x 2 = 14]

(a) Quadrature Phase Shift Keying

(b) Syndrome calculation method for error detection in cyclic codes

(c) Optical communication system and its componentthe

(d) Linear Block Codes

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